

Table 18 Medical air 400 kPa – design and diversified flows

Department	Design flow for each terminal unit (L/min)	Diversified flow Q (L/min)
In-patient accommodation (ward units):		
Single/multi-bed and treatment rooms ⁽¹⁾	20	$Q_w = 20 + [(n-1)10/4]$
Ward block/department	20	$Q_d = Q_w[1 + (nW-1)/2]$
Accident & emergency:		
Resuscitation room, per trolley space	40	$Q = 40 + [(n-1)20/4]$
Major treatment/plaster room, per trolley space	40	$Q = 40 + [(n-1)20/4]$
Post-anaesthesia recovery, per trolley space	40	$Q = 40 + [(n-1)40/4]$
Operating:		
Anaesthetic rooms	40	No additional flow included
Operating rooms	40	$Q = 40 + [(nT-1)40/4]$
Post-anaesthesia recovery	40	$Q = 40 + [(n-1)10/4]$
Maternity:		
LDRP rooms:	40	$Q = 40 + [(n-1)40/4]$
Baby ⁽²⁾	40	$Q = 40 + [(n-1)40/4]$
Operating suites:		
Anaesthetist	40	$Q = 40 + [(nS-1)10/4]$
Post-anaesthesia recovery	40	$Q = 40 + [(n-1)40/4]$
Neonatal unit (SCBU)	40	$Q = 40n$
Radiological:		
All anaesthetic and procedures rooms	40	$Q = 40 + [(n-1)40/4]$
Critical care areas⁽³⁾	80	$Q = 80 + [(n-1)80/2]$
High-dependency units	80	$Q = 80 + [(n-1)80/2]$
Renal	20	$Q = 20 + [(n-1)10/4]$
Oral surgery/orthodontic:		
Major dental/oral surgery rooms	40	$Q = 40 + [(n-1)40/2]$
All other departments	40	No additional flow allowance to be made
Equipment service rooms	40	No additional flow included

Notes:

1. It is assumed that a patient will use oxygen in a ward or in the treatment room.
2. Where two cot spaces have been provided in an LDRP room, assume only one will require medical air.
3. This diversified flow is also used for helium/oxygen mixture (see [paragraph 4.73](#)).

Surgical air 700 kPa

4.48 The pressure requirements of surgical tools are between 600 and 700 kPa and flows may vary between 200 and 350 L/min (STP; see [Table 19](#)). Most surgical tools are designed to operate within this pressure range. Higher pressures are likely to cause damage to tools. Inadequate tool performance, however, is likely to result from the lack of flow at the specified pressure.

4.49 The introduction of synthetic air (from on-site blending of oxygen and nitrogen) leads to the

possibility of using nitrogen as the power source for surgical tools.

4.50 The pipeline systems should be designed to provide a flow of 350 L/min at 700 kPa at the outlet from the terminal unit. Existing systems may not meet this requirement (but should be capable of delivering 250 L/min at the terminal unit).

Note

Some surgical tools require up to 500 L/min at up to 1400 kPa. These will require a separate supply, normally from cylinders.